

A Packing of 11 Four-Spaces in $\mathbb{R}^{10}$

Limin Ge*

October 9, 2026

Abstract

We give 11 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.639720. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.617891, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 11 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.639720 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 11$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.617891, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 11/10 = 2.640000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N11.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
263972087001650183497945920335546159857237327969581926339688978596954594
197062228324822197317170307865972641547991376061283264888894829934515175
9381340643484890799611996997569491524849918433677
```

denominator:

```
100000000000161053867106452138133633844587578584316055723267065257521539
294916147981039207538365508846466147110399379123381790117867256098081519
9605346137324691657792838734729206896324986591885
```

B Coordinates

11 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N11.txt`.

```
564311753761 164484608084 148777964930 -2142610729351912462539 -556517237082 -416652483467 -2182650551222829171190 -277820444835 379305708013 -380647544706
525870266805 465223838970 -407056572610 11697714027318695737752 221802044705 -512176496461 -1661772350436920651132 155945603071 698661073398 309804477635
158498366691 -146930267682 -353154480082 -48297441633275756114 -193702886220 -335618113754 5294321182476323669870 494702248944 60594105065 339307117637
-39142489696 259803378372 -124031544473 19885806056250617491874 -392333475177 -54827663523 -528085072016226556946 -154637374207 211652836916 -361976226764
72106985974 206090258013 615917853005 -14534000828 -316860169298 25260444769 309886538490 -13647671868201821829381 168738187768 -29840195130 371149399758
-354666736073 657969968695 -93494404360 12249232138341946940158 -160802844439 85015583300 -6615100203534082845 276560523686 52491248754 -339388373733
-358132981816 136604986485 61988351617 -21902837781-60929262886 218008776002 204212019110 25100673882 336798151878 548684806261 -300983761889 180129522746
-167318449203 -192760940517 -512183372688 283954725534228708901 -29935985450 -78491875031 52299280205265113948691 -301500701418 121796116909 727582192443
64583279283 -332245763675 18507012110 126191828786 -236404722477 94212296627 -630124116376 -960932125326478705202 -276403844937 314113780167 -260969392094
-294864770937 160694337887 -132118413676 -745217902354013363484 -575573642868 155579963505 360028771687672144320638 -167967509874 -393395622469 298134646528
355100458652 -108682575749 -425090398482 -3124551494533092269114 190555535313 -175741090718 7187268043974151769792 -268840109355 43676291309 388029229679
-135249697528 140456212101 259437301837 59890576291290148495678 -53101842076 579499963190 2077660821352246797374 31417689327 -54814221420 -63960789999
-282769074115 -252235040922 -425062995545 -1396141633324165388 625989310529 240637533626 103004258093213386072947 304460042121 47887392433 126264185720
-387417106717 -154978105470 -269877214858 -14065103853227184278 902133702608 77987572113 -12712813761388905551476 -76127951467 -565779249082 40124047708
42135679725 696408290456 -221884186440 -24749304885705376481562 -222391376034 25072696416 -31410355629710263789187 -349535102471 182921422175 -59474809164
-226518398671 -75715351838 -228667939374 3885290634426108739707 193270295539 -296669008149 3099350033329120928738 -457713769566 -529504554549 -324716360441
-259785881486 -492573277924 351035712652 -786710541332439436547 -153561881898 -30199150641 -435139397812772414049 -411771769298 -573853621325 295892302397
220153759519 -193699362959 -291756136579 299131474372589062439 116937253739 236846139084 -1194583747938856109974 196539514819 -265767883298 -211022791985
-501433737504 137486660307 -363178335834 21421647344980449164941 -172268057906 -106586691491 572891004207504596770 -320380645475 461321687891 -220241005143
-452860512886 301747643817 238139331390 -45965639957233331025981 -163576670440 599893380584 -136637201234565840646 -44993952311 -439898645220 -482088465269
5279726396 -17735087090 -435772287500 -79805731683 -49076035308 59630520722 407283294353 42855119732 -209219705810 -106272209580 -312213393927 -342433905299
187653610420 -572270010221 53808820384 24421170659381737887289 28262911451 557811295662 448449723634 -110565306701 -588644958851 -117563894713 215590273937
297964695886 118427781824 199549730075 -60164743593883441713412 36051025082 -1673767167 -188900884119-292062788822 242876792630 -136870237734 -256512266744
-38862575867 -472386811977 -455960911207 -50074032408442837676 -284974505333 -120173015043 32931556676122516979293 -120756625028 36755013395 -437581416463
458278587920 29882126004 11727764659 340477892088 -63862340114 -28012706857 614894504547 9285642969 48954641224 332310212222 -211952322629 393474237834
-408341590263 424493408595 -348912755680 21553861897314161723817 348757285448 341447098511 -4324084779398621253688 381247910474 -132001071860 105657831591
300099794638 143535455068 225994093807 -10264840948271457655075 -385966395998 -290651459242 -359749924066887505 262936707623 -10845231339 -553076149488
26502906898 291711391440 310978247044 -216360997322327555007548 463119074425 176237426896 19154787754 -80955217185 -301000196587 43287457775 -434027167101
570023859045 357360361827 -538309905518 -144550030594106182825 -461211743018 194988821381 -631176195672929648846 21980584996 -323169239547 24032094306
-286379779326 143269243359 7118320373 -318325875046705580764861 229924856997 -251795190335 -1809352427956771486587 585215822628 72783745765 85715799443
-182216926615 -25516096538 95028784728 -2054659968748887344364 -199004204959 47704899103 242369350138793623861458 -37860748151 248267366681 144105983878
-72743341103 -118950591094 183538670056 5212818219478404848048 -311950211113 517187876299 25144214586379050477111 -74323543605 111789326972 -192256792658
419824358960 -514163670797 356025314373 18684854722096999903077 -174431992213 64487650946 147017070091-66165943556 -550908802649 -173784802579 -321898463658
9747464562 22076289680 329694192179 -379893210295 -273006961242 433109718673 -177306381030 -409582040631323615013 -183629115968 380054735223 302623275831
584921298292 99364272460 320755218525 -1287796929624967065788 -123722034662 -262557775537 25579527934-52587081069 332853741784 -479346032102 -229123404699
142687012704 188863517530 239389703508 239439363144-458896383269 10063121233 -442566584612 24262580594251551731 205750490635 640096734151 -438607158836
-441393168339 -527614409899 114349847805 -2781177123211888368973 -558562117342 -140164998978 301721303753
```